CS23532 - COMPUTER NETWORKS LABORATORY

EXPERIMENT 5

AIM: Experiments on Packet capture tool: Wireshark. To understand the features of wireshark as a packet capture tool and understand encapsulation of information at various layers of a Protocol stack.

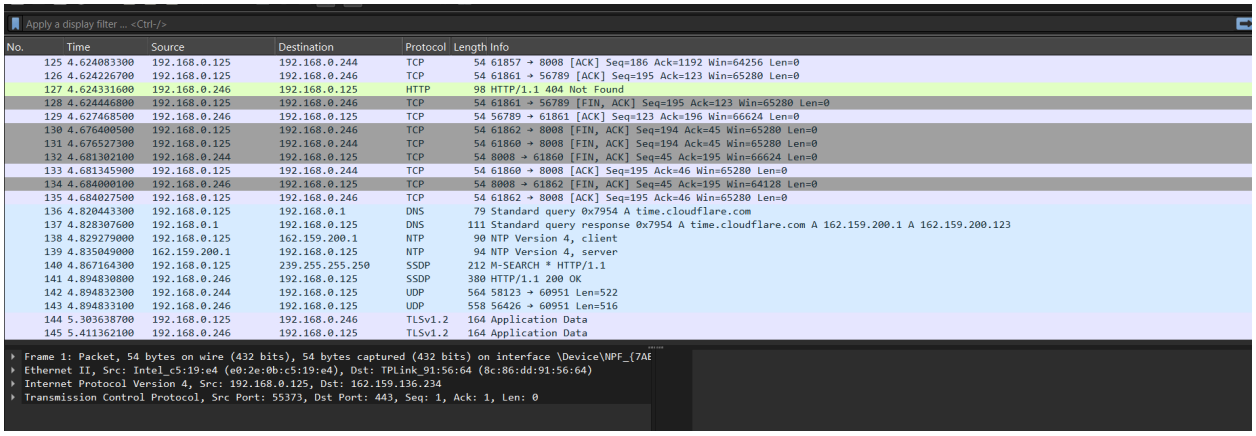
DESCRIPTION:

- **Packet Sniffer:** A passive program that sniffs messages being sent/received from/by your computer. It never sends packets itself, has no packets addressed to it, and receives a copy of all packets (sent/received). It stores and displays the contents of the various protocol fields in the messages.
- **Wireshark:** Wireshark, a network analysis tool formerly known as Ethereal, captures packets in real time and display them in human-readable format. Wireshark includes filters, color coding, and other features that let you dig deep into network traffic and inspect individual packets. You can use Wireshark to inspect a suspicious program's network traffic, analyze the traffic flow on your network, or troubleshoot network problems.

STEP-BY-STEP PROCEDURE

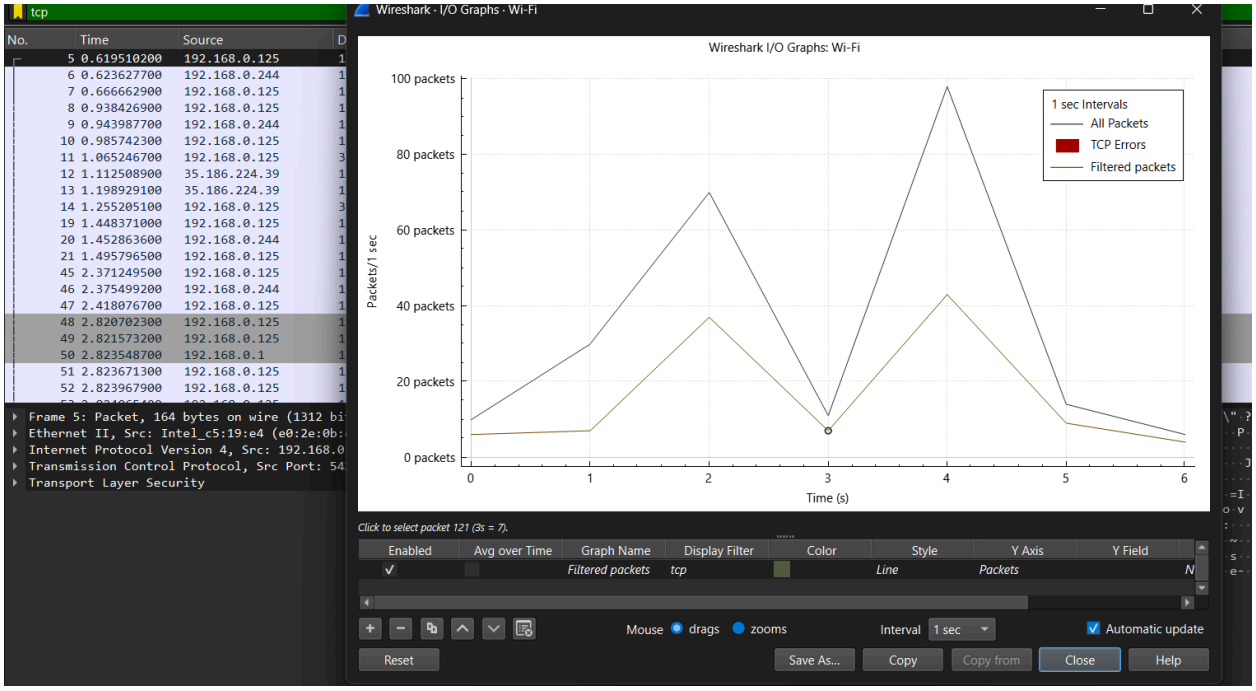
Initial Task: Capture 100 packets from the Ethernet: IEEE 802.3 LAN Interface and save it

1. Select Local Area Connection in Wireshark.
2. Go to capture → option.
3. Select stop capture automatically after 100 packets.
4. Then click Start capture.
5. Save the packets.



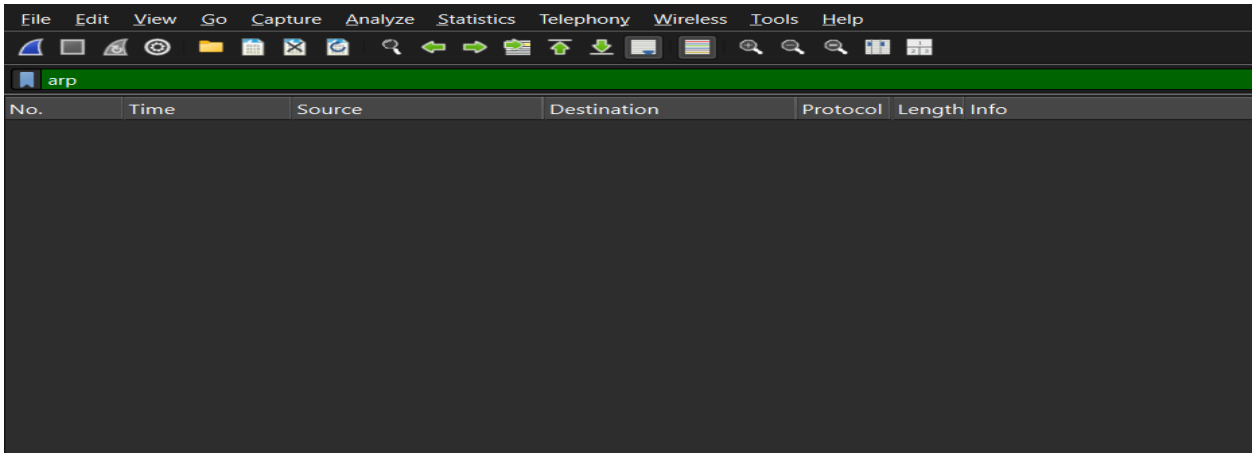
Task 1: Create a Filter to display only TCP/UDP packets, inspect the packets and provide the flow graph

1. Select Local Area Connection in Wireshark.
2. Go to capture → option.
3. Select stop capture automatically after 100 packets.
4. Then click Start capture.
5. Search TCP packets in search bar.
6. To see flow graph click Statistics → Flow graph.
7. Save the packets.



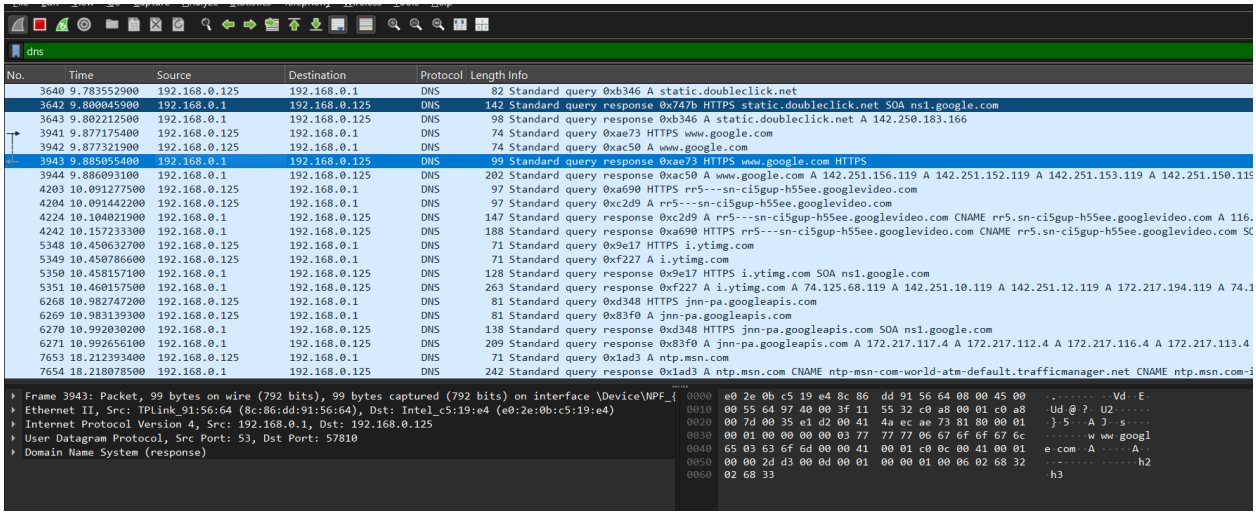
Task 2: Create a Filter to display only ARP packets and inspect the packets

1. Go to capture → option.
2. Select stop capture automatically after 100 packets.
3. Then click Start capture.
4. Search ARP packets in search bar.
5. Save the packets.



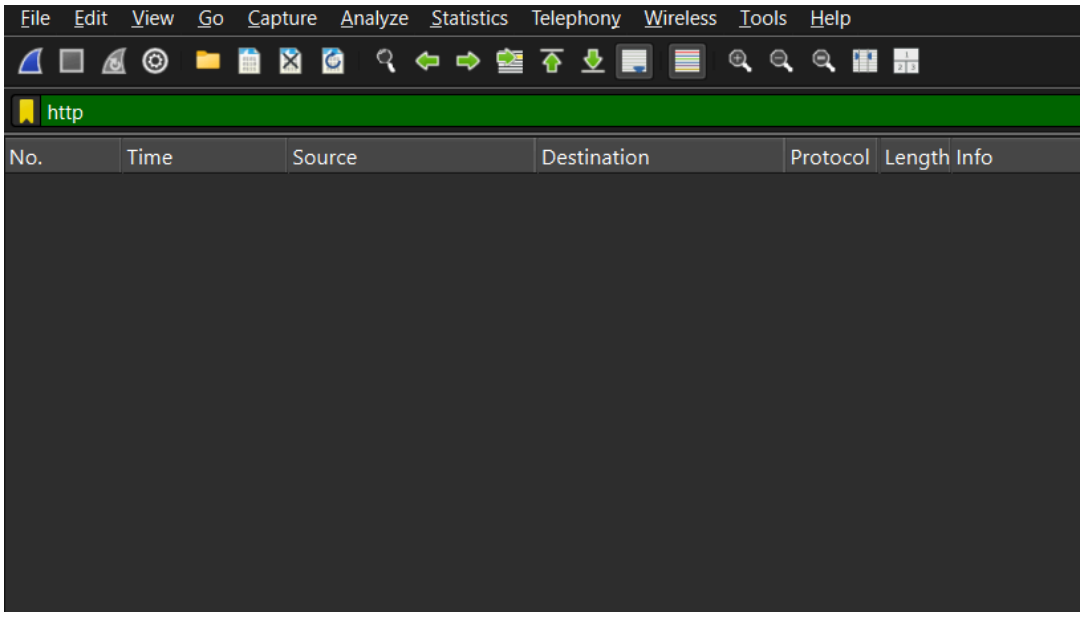
Task 3: Create a Filter to display only DNS packets and provide the flow graph

1. Go to capture → option.
2. Select stop capture automatically after 100 packets.
3. Then click Start capture.
4. Search DNS packets in search bar.
5. To see flow graph click Statistics → Flow graph.
6. Save the packets.



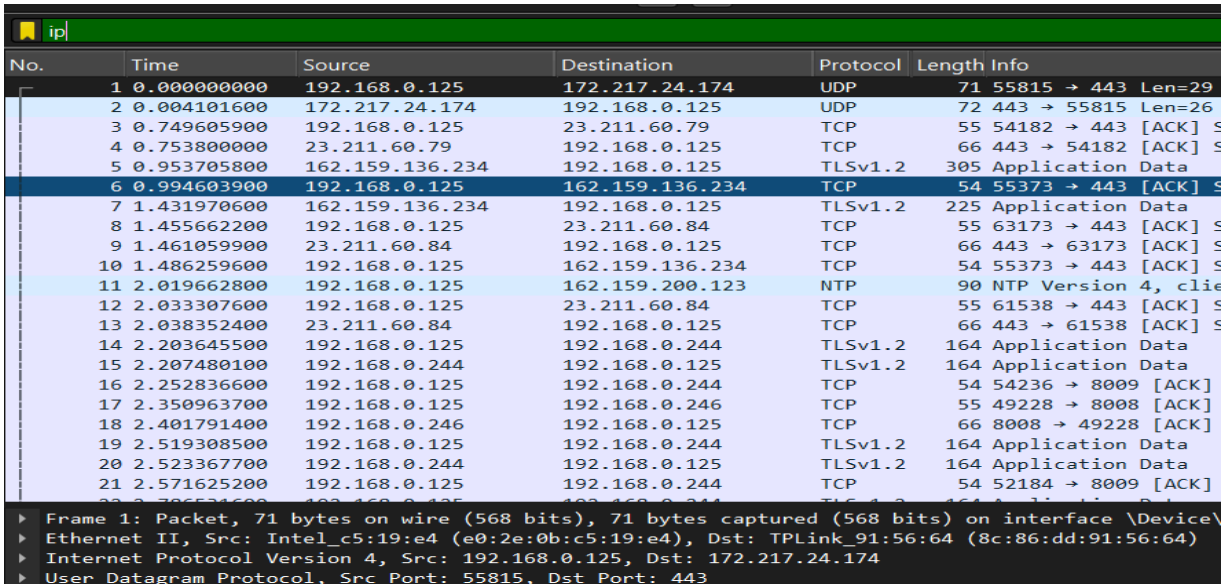
Task 4: Create a Filter to display only HTTP packets and inspect the packets

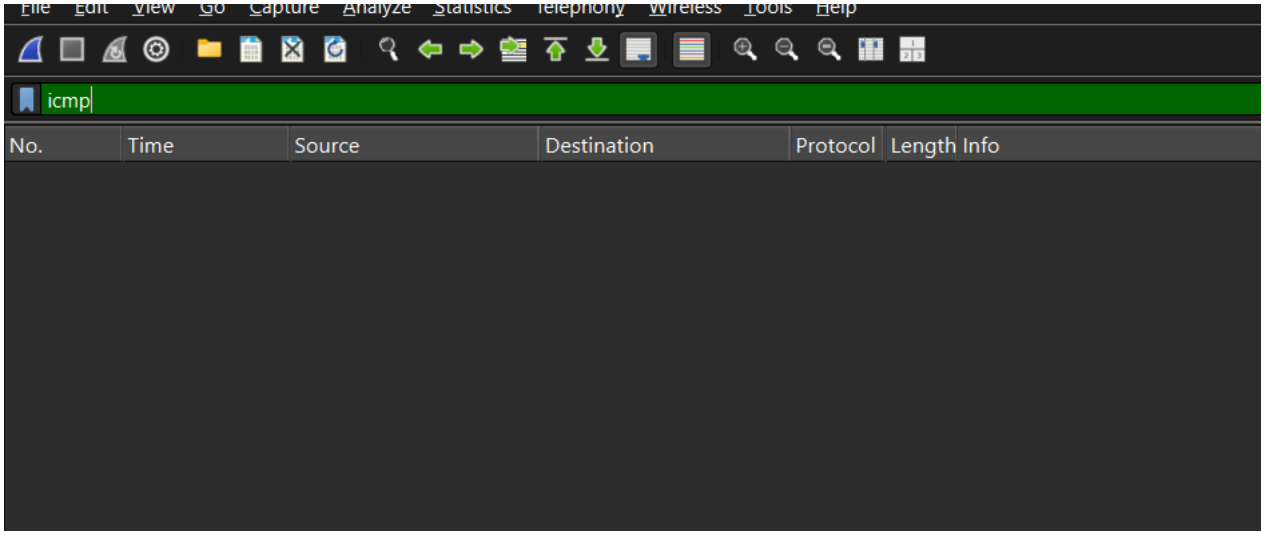
1. Select Local Area Connection in Wireshark.
2. Go to capture → option.
3. Select stop capture automatically after 100 packets.
4. Then click Start capture.
5. Search HTTP packets in search bar.
6. Save the packets.



Task 5: Create a Filter to display only IP/ICMP packets and inspect the packets

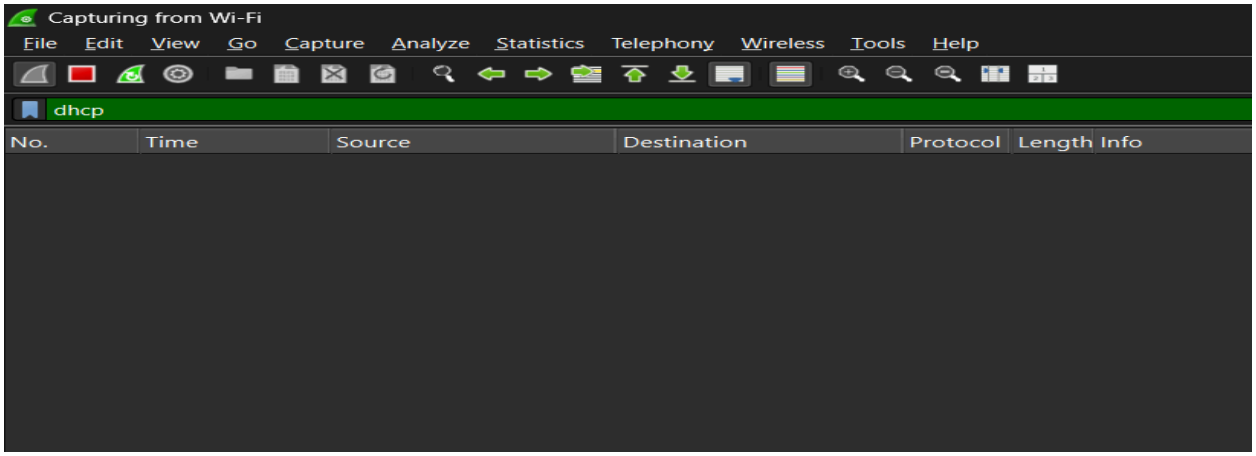
1. Select Local Area Connection in Wireshark.
2. Go to capture → option.
3. Select stop capture automatically after 100 packets.
4. Then click Start capture.
5. Search ICMP/IP packets in search bar.
6. Save the packets.





Task 6: Create a Filter to display only DHCP packets and inspect the packets

1. Select Local Area Connection in Wireshark.
2. Go to capture → option.
3. Select stop capture automatically after 100 packets.
4. Then click Start capture.
5. Search DHCP packets in search bar.
6. Save the packets.



Result:

Thus the experiment to understand the features of wireshark as a packet capture tool and understand encapsulation of information at various layers of a Protocol stack has been studied and executed successfully.